

Ultimate DSA Placement Dashboard - SRS

Version 2

Volume 2: Master Problem Database, Revision Engine, and Data Model

1. Master Problem Database Columns

Column	Purpose
ID	Unique numeric identifier
Problem Name	Canonical problem title
Primary Topic	Arrays, Trees, etc.
Pattern	Sliding Window, Kadane, DFS, etc.
Difficulty	Easy/Medium/Hard
ROI	1-10
Frequency	★★★★★ to ★☆☆☆☆
Striver	Hyperlink
LeetCode	Hyperlink
GFG	Hyperlink
Attempt Date	Latest attempt
Time	Minutes
Status	Green/Yellow/Red
Hint Used	Yes/No
Confidence	1-5
Revision Count	Integer
Next Revision	Auto date
Notes	Mistakes/Observation

2. Problem Categories

Maintain approximately 300-350 curated problems. Categorize by Arrays, Hashing, Strings, Binary Search, Sliding Window, Two Pointer, Linked List, Stack, Queue, Trees, BST, Heap, Greedy, Graphs, DP, Bit Manipulation and Miscellaneous.

3. Pattern Library

Every problem must be tagged with one or more patterns such as Sliding Window, Prefix Sum, Binary Search on Answer, Kadane, Monotonic Stack, DFS, BFS, Tree Traversal, Interval Greedy, Hash Map, Backtracking, Union Find.

4. Revision Engine

Green: review after ~30 days.

Yellow: review after 7 days.

Red: review after 2 days.

Generate today's revision queue by sorting overdue problems first, then Yellow, then Red, while mixing topics.

5. Mixed Revision Rule

Never revise one topic exclusively. Every revision session should include a balanced mix of Arrays, Binary Search, Strings, Linked List, Trees, Stack, Greedy and other topics.

6. Data Validation

Dropdowns: Difficulty, Status, Hint Used, Confidence.

Hyperlinks must open Striver, LeetCode and GFG resources directly.

7. Success Criteria

The workbook should answer:

- What should I revise today?
- Which patterns are weak?
- Which companies am I ready for?
- Which problems keep recurring?
- Which topics have been forgotten?